

Rotation motion in a nutshell

Rotational Kinematics

<http://hyperphysics.phy-astr.gsu.edu/hbase/circ.html>

In the simplest kind of rotation, points on a rigid object move on circular paths around an ***axis of rotation***.



We can define:

Angular displacement (radians)

Angular speed/velocity (rpm)

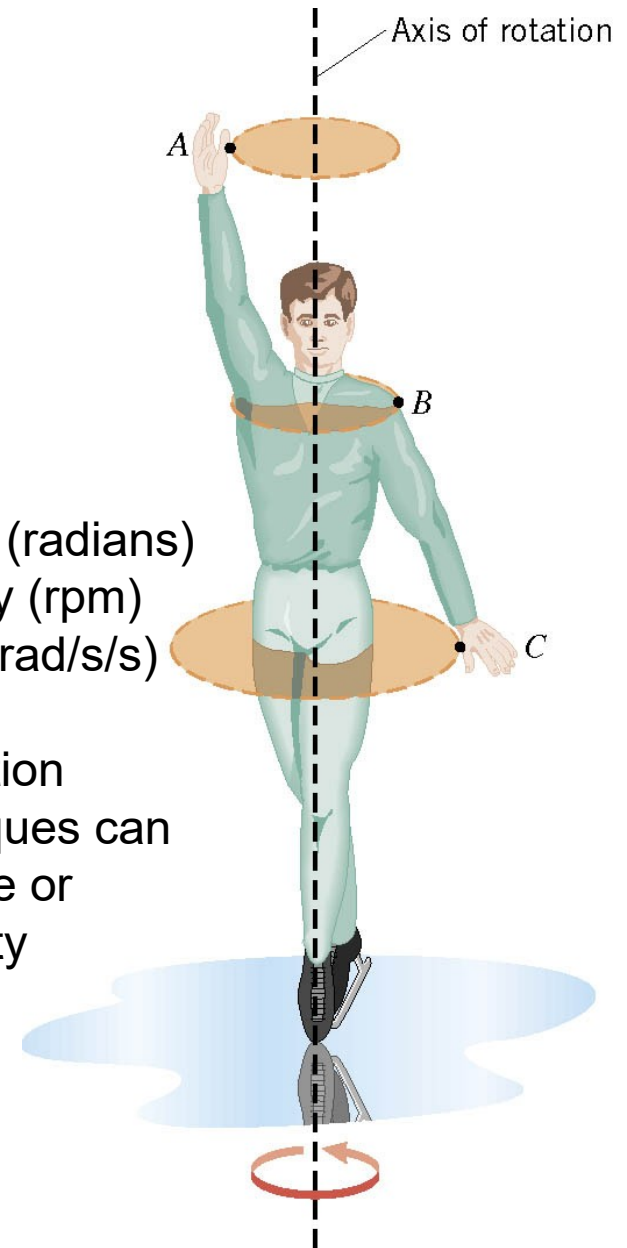
Angular acceleration (rad/s/s)

Velocity and acceleration

Are acceleration. Torques can

Change the magnitude or

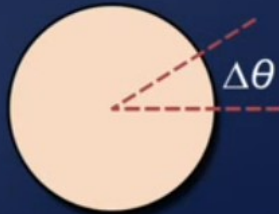
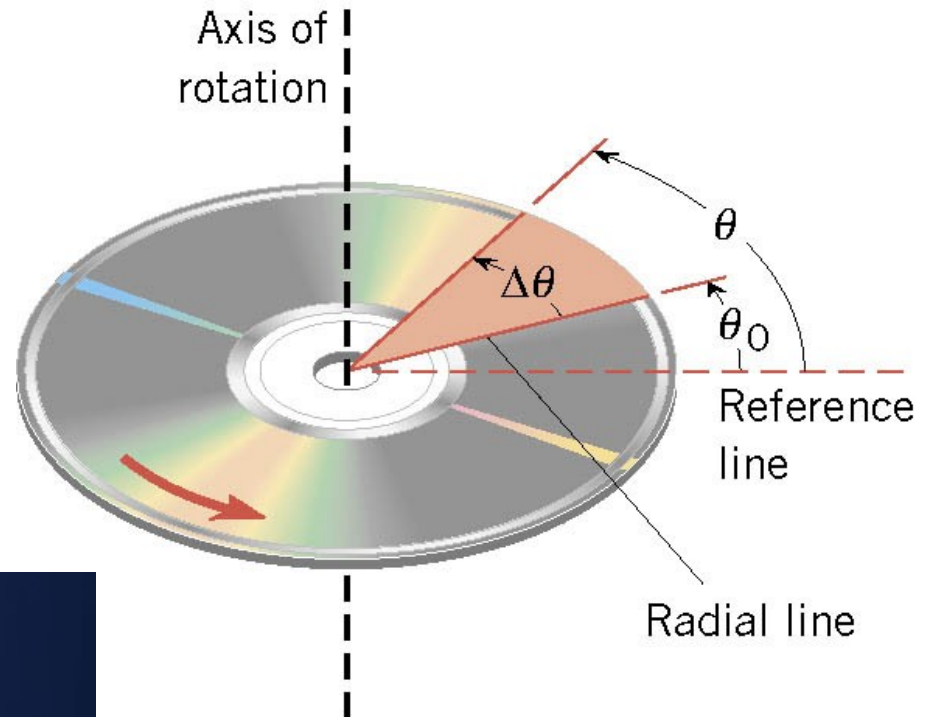
Direction of the velocity



**Rotational analog of displacement Motion →
Angular Displacement**

The angle through which the object rotates is called the **angular displacement**.

$$\Delta\theta = \theta - \theta_0$$



Displacement: $\Delta x \Rightarrow$ Angular displacement $\Delta\theta$

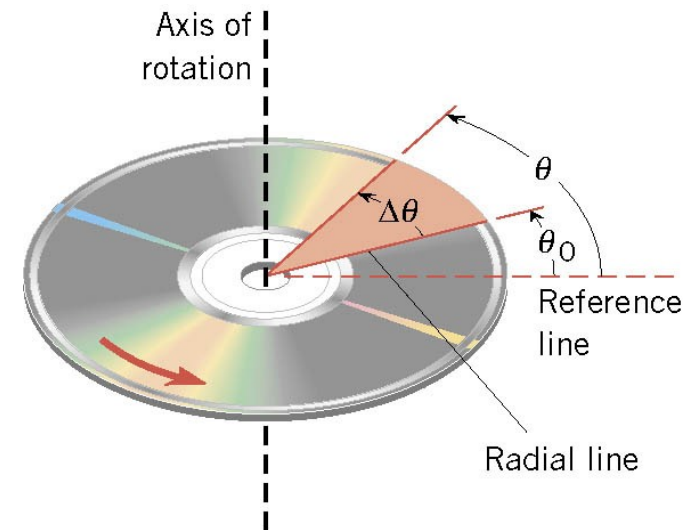
- Revolutions
- Degrees
- Radians

Rotational Motion and Angular Displacement

DEFINITION OF ANGULAR DISPLACEMENT

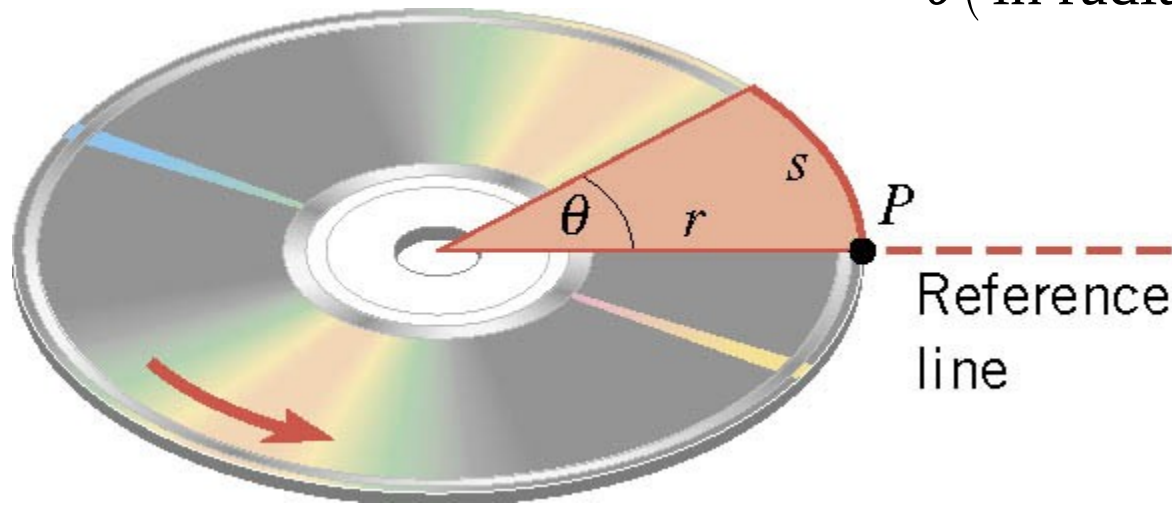
When a rigid body rotates about a fixed axis, the angular displacement is the angle swept out by a line passing through any point on the body and intersecting the axis of rotation perpendicularly.

By convention, the angular displacement is positive if it is counterclockwise and negative if it is clockwise.



SI Unit of Angular Displacement: radian (rad)

8.1 Rotational Motion and Angular Displacement



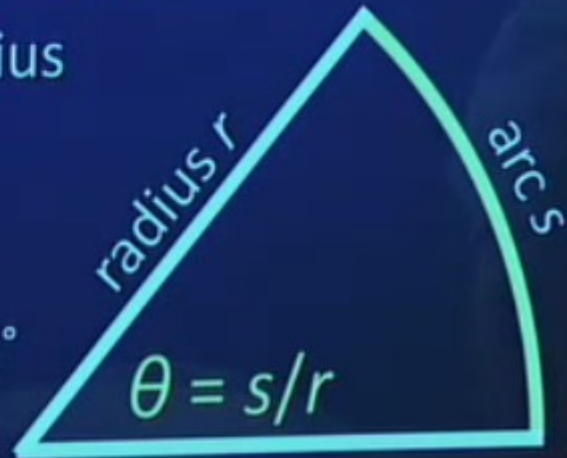
$$\theta(\text{ in radians }) = \frac{\text{Arc length}}{\text{Radius}} = \frac{s}{r}$$

For a full revolution:

$$\theta = \frac{2\pi r}{r} = 2\pi \text{ rad} \quad \longrightarrow \quad 2\pi \text{ rad} = 360^\circ$$

What's This Radian?

- Radian: the “natural” measure of angle
 - Also the official SI unit of angle
 - Ratio of arc length to radius
 - Dimensionless
 - 2π radians in a full circle
 - $1 \text{ radian} = 360^\circ/2\pi \approx 57.3^\circ$

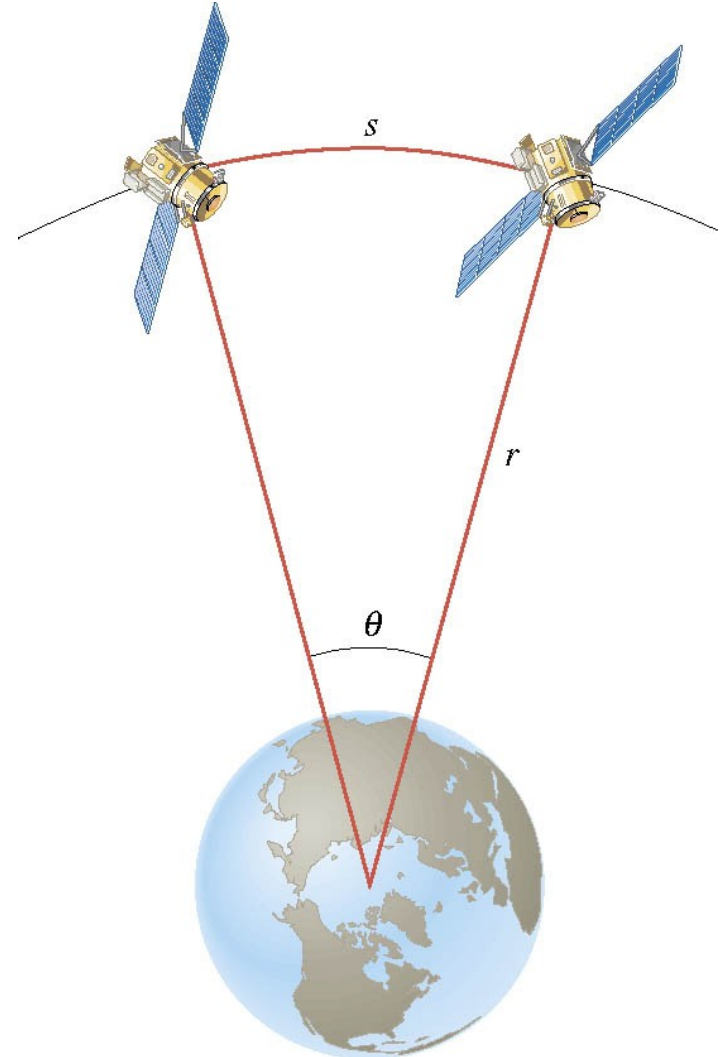


8.1 Rotational Motion and Angular Displacement

Example 1 Adjacent Synchronous Satellites

Synchronous satellites are put into an orbit whose radius is $4.23 \times 10^7 \text{ m}$.

If the angular separation of the two satellites is 2.00 degrees, find the arc length that separates them.



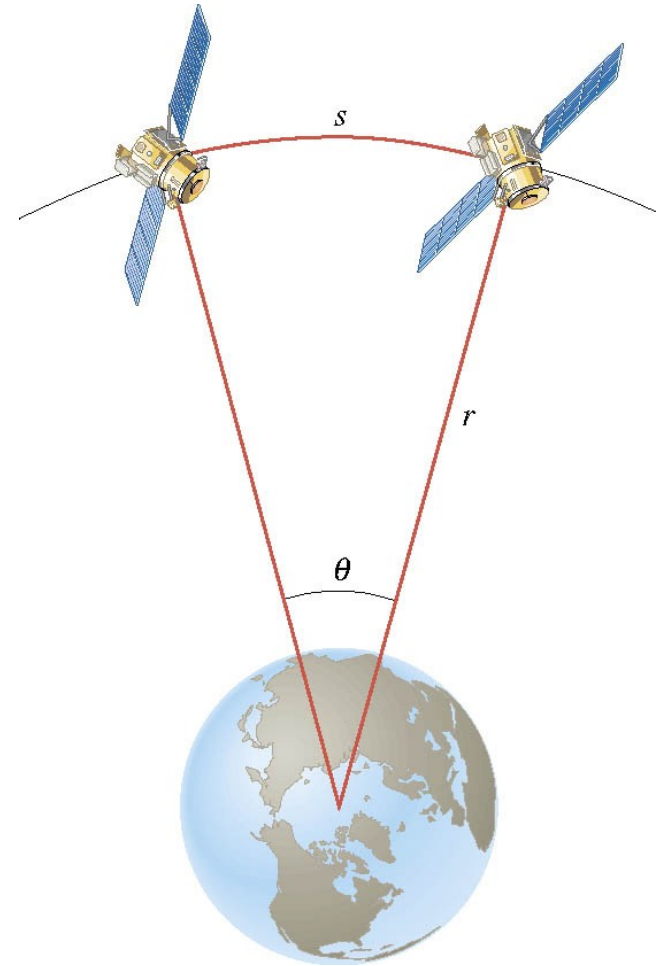
8.1 Rotational Motion and Angular Displacement

$$\theta(\text{ in radians }) = \frac{\text{Arc length}}{\text{Radius}} = \frac{s}{r}$$

$$2.00 \text{ deg} \left(\frac{2\pi \text{ rad}}{360 \text{ deg}} \right) = 0.0349 \text{ rad}$$

$$s = r\theta = (4.23 \times 10^7 \text{ m})(0.0349 \text{ rad})$$

$$1.48 \times 10^6 \text{ m} \text{ (920 miles)}$$

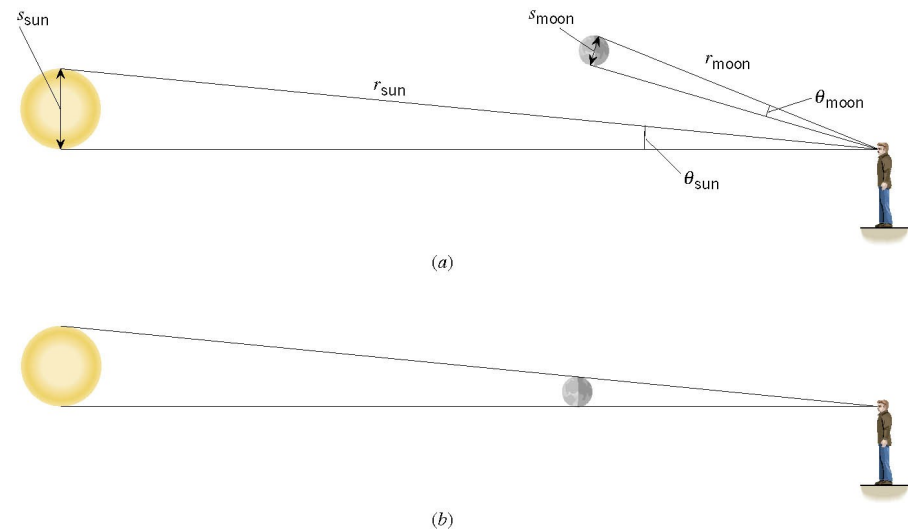


8.1 Rotational Motion and Angular Displacement

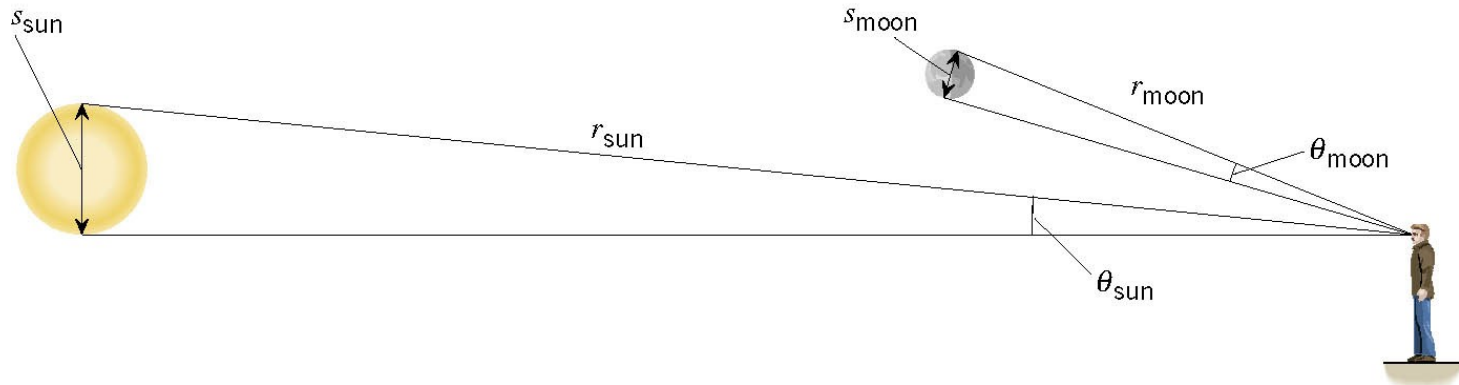
Conceptual Example 2 A Total Eclipse of the Sun

The diameter of the sun is about 400 times greater than that of the moon. By coincidence, the sun is also about 400 times farther from the earth than is the moon.

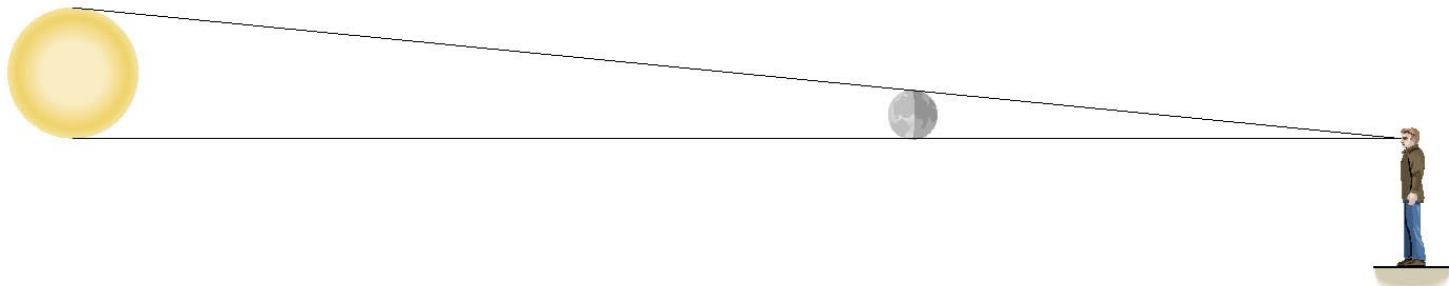
For an observer on the earth, compare the angle subtended by the moon to the angle subtended by the sun and explain why this result leads to a total solar eclipse.



8.1 Rotational Motion and Angular Displacement



(a)

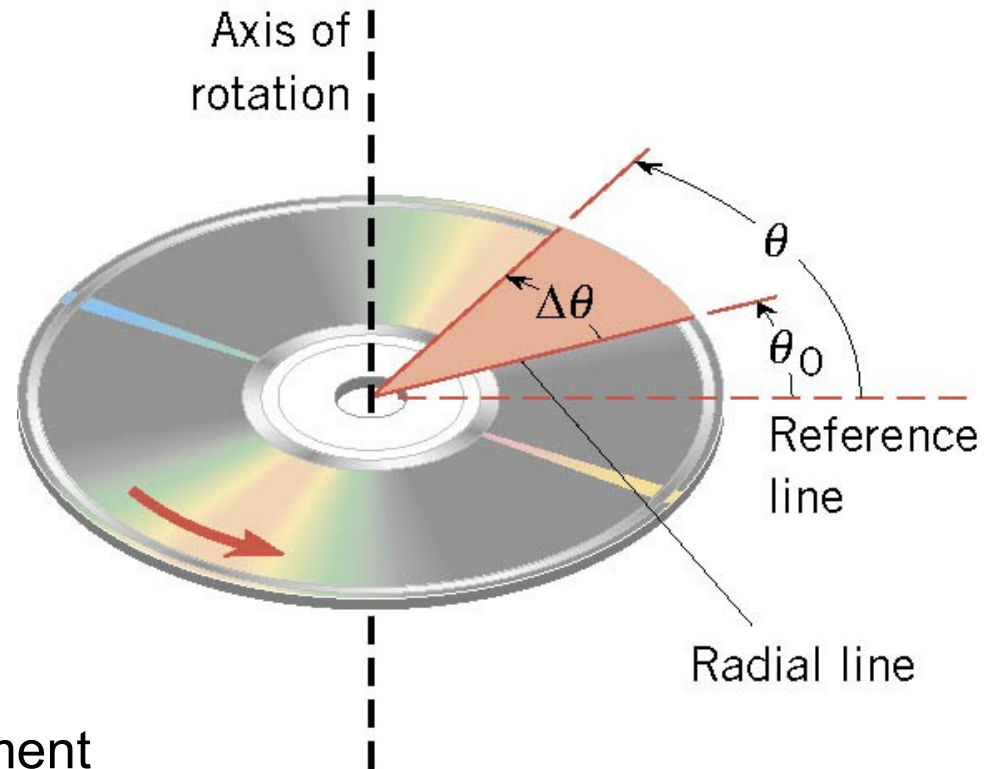


(b)

$$\theta(\text{ in radians }) = \frac{\text{Arc length}}{\text{Radius}} = \frac{s}{r}$$

8.2 Angular Velocity and Angular Acceleration

$$\Delta\theta = \theta - \theta_0$$



How do we describe the rate at which the angular displacement is changing?

Rotational analog of velocity = Angular Velocity

DEFINITION OF AVERAGE ANGULAR VELOCITY

$$\text{Average angular velocity} = \frac{\text{Angular displacement}}{\text{Elapsed time}}$$

$$\bar{\omega} = \frac{\theta - \theta_o}{t - t_o} = \frac{\Delta\theta}{\Delta t}$$

SI Unit of Angular Velocity: radian per second (rad/s)

Velocity: $v = \Delta x / \Delta t \Rightarrow$ Angular velocity $\omega = \Delta\theta / \Delta t$

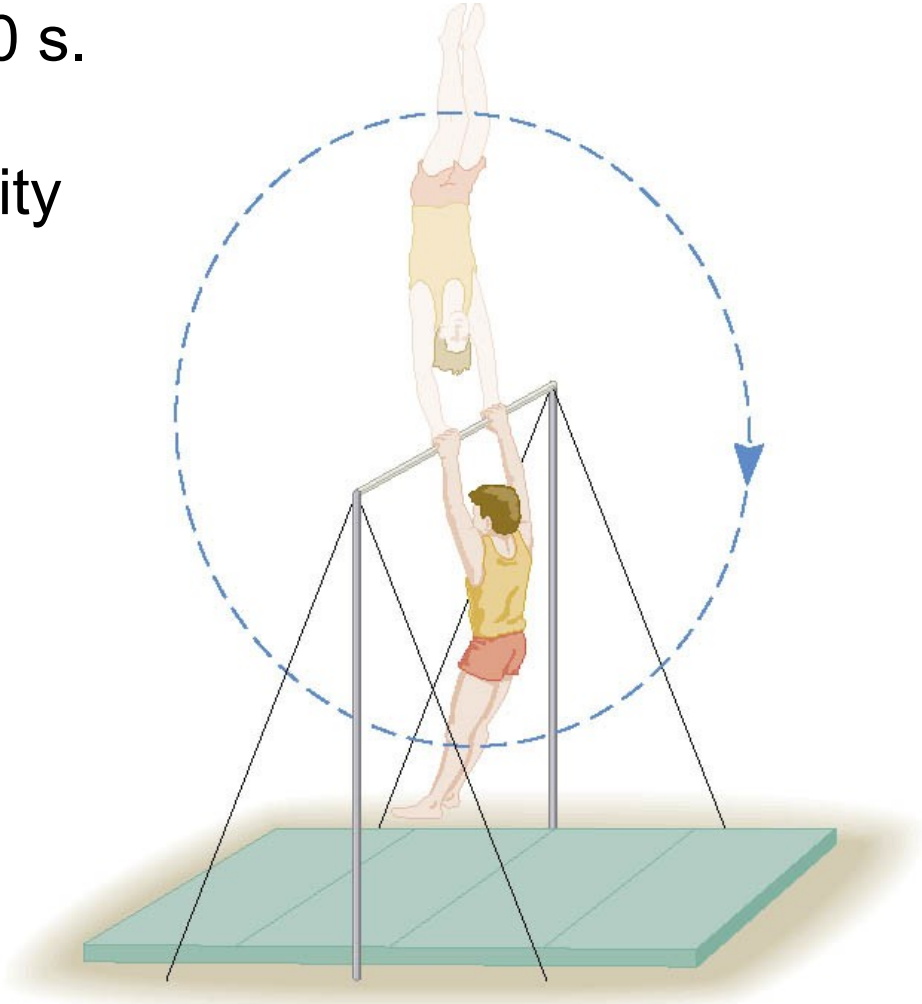
- Revolutions per minute (rpm)
- Degrees per second
- Radians per second (s^{-1})

8.2 *Angular Velocity and Angular Acceleration*

Example 3 Gymnast on a High Bar

A gymnast on a high bar swings through two revolutions in a time of 1.90 s.

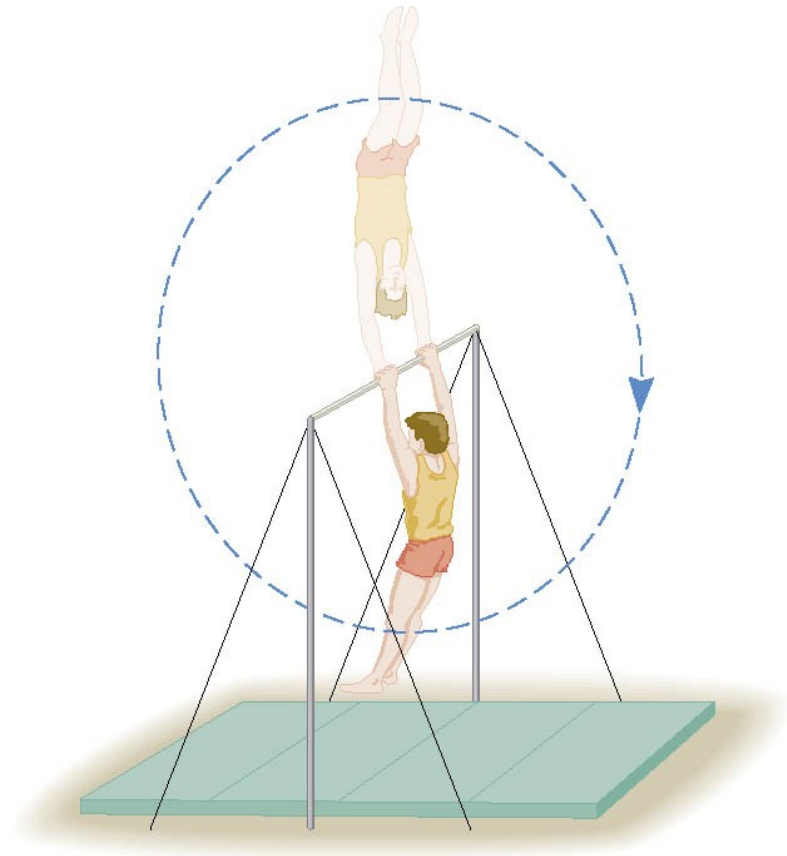
Find the average angular velocity of the gymnast.



8.2 Angular Velocity and Angular Acceleration

$$\Delta\theta = -2.00 \text{ rev} \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) = -12.6 \text{ rad}$$

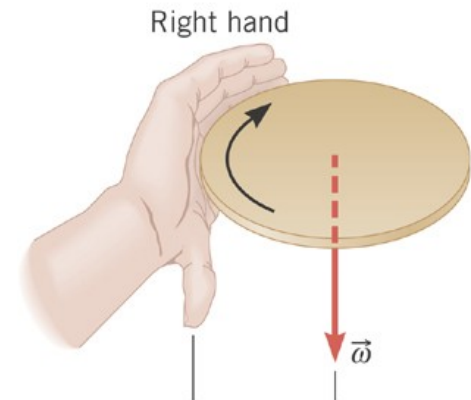
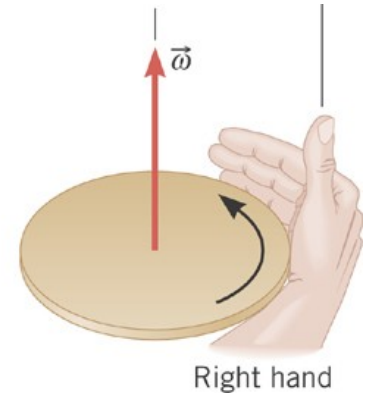
$$\bar{\omega} = \frac{-12.6 \text{ rad}}{1.90 \text{ s}} = -6.63 \text{ rad/s}$$



8.7 The Vector Nature of Angular Variables

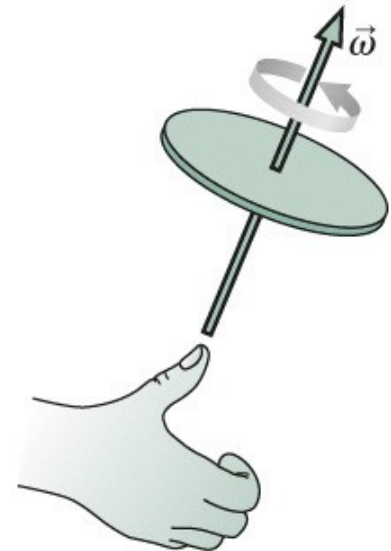
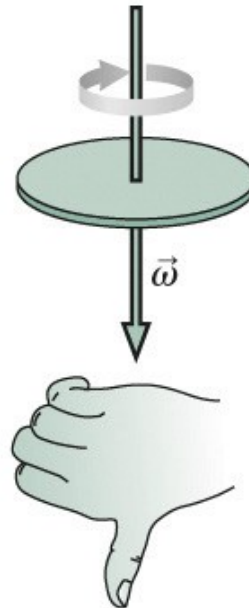
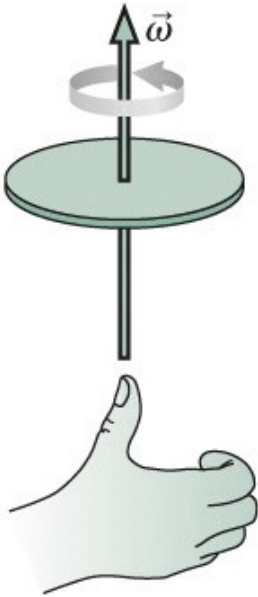
Right-Hand Rule: Grasp the axis of rotation with your right hand, so that your fingers circle the axis in the same sense as the rotation.

Your extended thumb points along the axis in the direction of the angular velocity.



Direction of the Angular Velocity Vector

- The direction of angular velocity is given by the **right-hand rule**.
 - Curl the fingers of your right hand in the direction of rotation, and your thumb points in the direction of the angular velocity vector $\vec{\omega}$.



Rotational analog of acceleration
Angular Acceleration

INSTANTANEOUS ANGULAR VELOCITY

$$\omega = \lim_{\Delta t \rightarrow 0} \bar{\omega} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t}$$

Acceleration: $a = \Delta v / \Delta t \Rightarrow$ Angular acceleration $\alpha = \Delta \omega / \Delta t$

- rpm/second
- (degrees/second)/second
- radians/second² (s⁻²)

8.2 Angular Velocity and Angular Acceleration

Changing angular velocity means that an **angular acceleration** is occurring.

DEFINITION OF AVERAGE ANGULAR ACCELERATION

Average angular acceleration = $\frac{\text{Change in angular velocity}}{\text{Elapsed time}}$

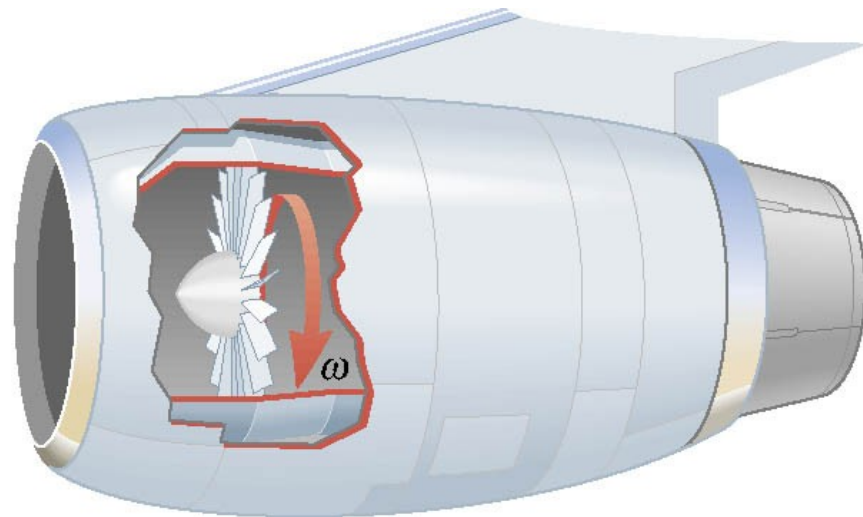
$$\bar{\alpha} = \frac{\omega - \omega_o}{t - t_o} = \frac{\Delta\omega}{\Delta t}$$

SI Unit of Angular acceleration: radian per second squared (rad/s²)

8.2 Angular Velocity and Angular Acceleration

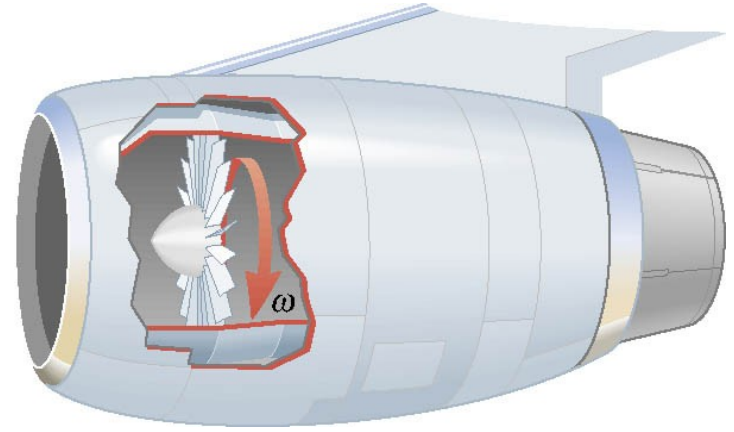
Example 4 A Jet Revving Its Engines

As seen from the front of the engine, the fan blades are rotating with an angular speed of -110 rad/s . As the plane takes off, the angular velocity of the blades reaches -330 rad/s in a time of 14 s .



Find the angular acceleration, assuming it to be constant.

8.2 Angular Velocity and Angular Acceleration



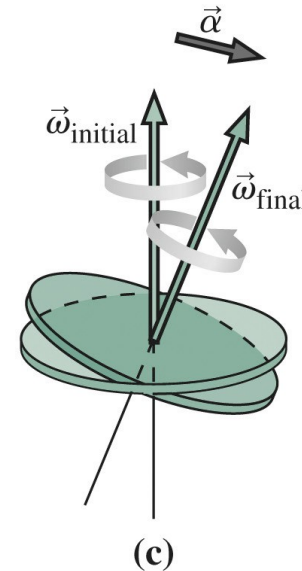
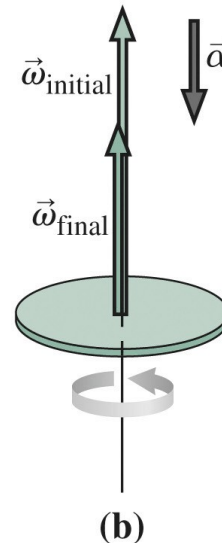
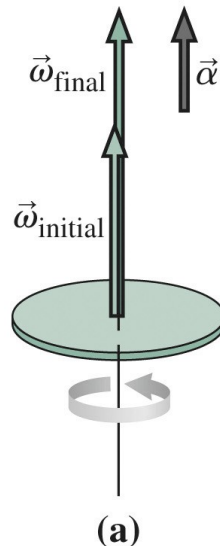
$$\bar{\alpha} = \frac{(-330 \text{ rad/s}) - (-110 \text{ rad/s})}{14 \text{ s}} = -16 \text{ rad/s}^2$$

Direction of the Angular Acceleration

- Angular acceleration points in the direction of the change in the angular velocity $\Delta\vec{\omega}$:

$$\vec{\alpha} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{\omega}}{\Delta t} = \frac{d\vec{\omega}}{dt}$$

- The change can be in the same direction as the angular velocity, increasing the angular speed.
- The change can be opposite the angular velocity, decreasing the angular speed.
- Or it can be in an arbitrary direction, changing the direction and speed as well.



8.3 *The Equations of Rotational Kinematics*

Recall the equations of kinematics for constant acceleration.

Five kinematic variables:

$$v = v_o + at$$

1. displacement, x

$$x = \frac{1}{2} (v_o + v) t$$

2. acceleration (constant), a

3. final velocity (at time t), v

$$v^2 = v_o^2 + 2ax$$

4. initial velocity, v_o

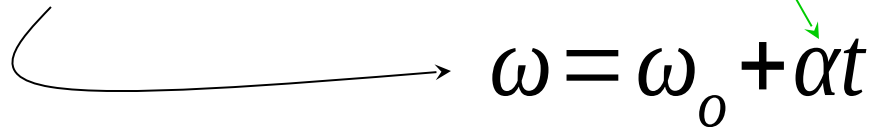
5. elapsed time, t

$$x = v_o t + \frac{1}{2} at^2$$

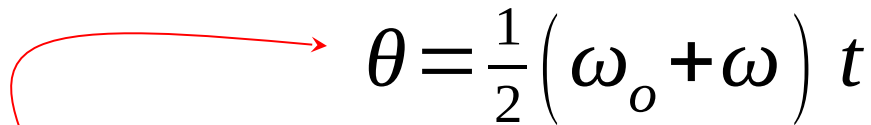
8.3 The Equations of Rotational Kinematics

The equations of rotational kinematics for constant angular acceleration:

ANGULAR VELOCITY


$$\omega = \omega_o + \alpha t$$

ANGULAR ACCELERATION


$$\theta = \frac{1}{2} (\omega_o + \omega) t$$

TIME

ANGULAR DISPLACEMENT

$$\omega^2 = \omega_o^2 + 2 \alpha \theta$$

$$\theta = \omega_o t + \frac{1}{2} \alpha t^2$$

8.3 The Equations of Rotational Kinematics

**Table 8.2 Symbols Used
in Rotational and Linear Kinematics**

Rotational Motion	Quantity	Linear Motion
θ	Displacement	x
ω_0	Initial velocity	v_0
ω	Final velocity	v
α	Acceleration	a
t	Time	t

**Table 8.1 The Equations of Kinematics
for Rotational and Linear Motion**

Rotational Motion ($\alpha = \text{constant}$)		Linear Motion ($a = \text{constant}$)	
$\omega = \omega_0 + \alpha t$	(8.4)	$v = v_0 + at$	(2.4)
$\theta = \frac{1}{2}(\omega_0 + \omega)t$	(8.6)	$x = \frac{1}{2}(v_0 + v)t$	(2.7)
$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$	(8.7)	$x = v_0 t + \frac{1}{2}at^2$	(2.8)
$\omega^2 = \omega_0^2 + 2\alpha\theta$	(8.8)	$v^2 = v_0^2 + 2ax$	(2.9)

8.3 *The Equations of Rotational Kinematics*

Reasoning Strategy

1. Make a drawing.
2. Decide which directions are to be called positive (+) and negative (-).
3. Write down the values that are given for any of the five kinematic variables.
4. Verify that the information contains values for at least three of the five kinematic variables. Select the appropriate equation.
5. When the motion is divided into segments, remember that the final angular velocity of one segment is the initial velocity for the next.
6. Keep in mind that there may be two possible answers to a kinematics problem.

8.3 *The Equations of Rotational Kinematics*

Example 5 Blending with a Blender

The blades are whirling with an angular velocity of $+375 \text{ rad/s}$ when the “puree” button is pushed in.

When the “blend” button is pushed, the blades accelerate and reach a greater angular velocity after the blades have rotated through an angular displacement of $+44.0 \text{ rad}$.

The angular acceleration has a constant value of $+1740 \text{ rad/s}^2$.

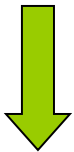
Find the final angular velocity of the blades.



8.3 The Equations of Rotational Kinematics

θ	α	ω	ω_o	t
+44.0 rad	+1740 rad/s ²	?	+375 rad/s	

$$\omega^2 = \omega_o^2 + 2 \alpha \theta$$



$$\omega = \sqrt{\omega_o^2 + 2 \alpha \theta}$$

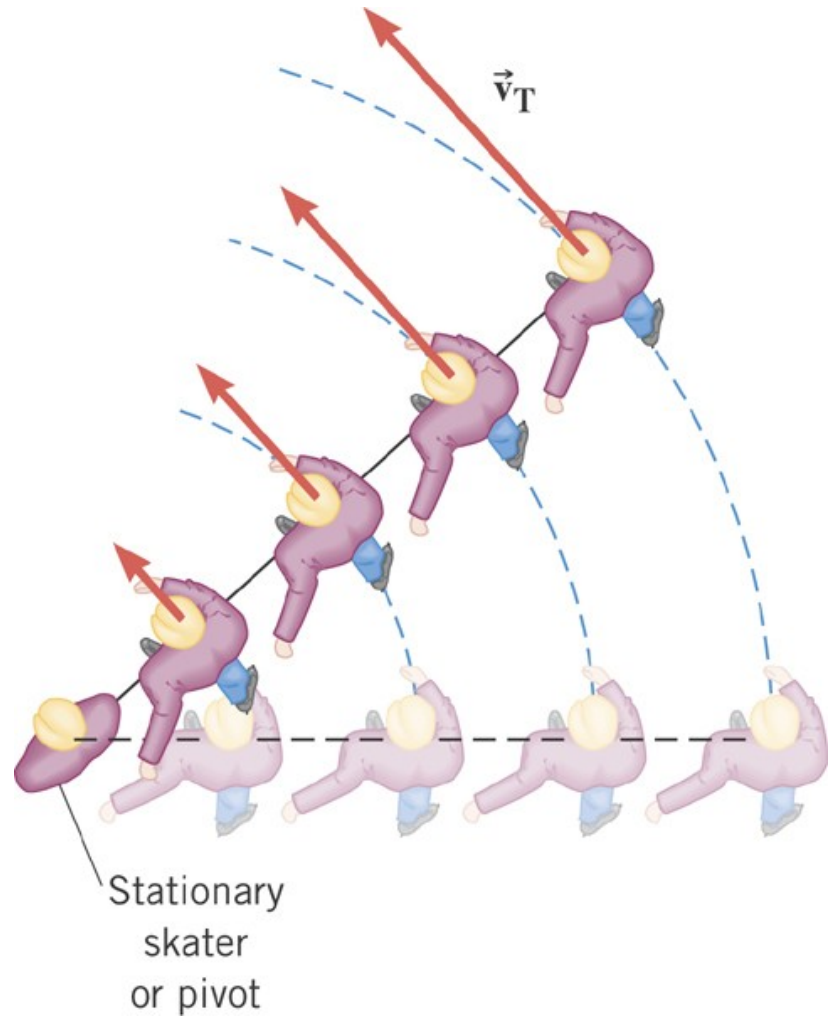
$$\therefore \sqrt{(375 \text{ rad/s})^2 + 2(1740 \text{ rad/s}^2)(44.0 \text{ rad})} = +542 \text{ rad/s}$$



8.4 Angular Variables and Tangential Variables

$\vec{v}_T$ = tangential velocity

v_T = tangential speed



<https://www.youtube.com/watch?v=5YODez4-qbU>

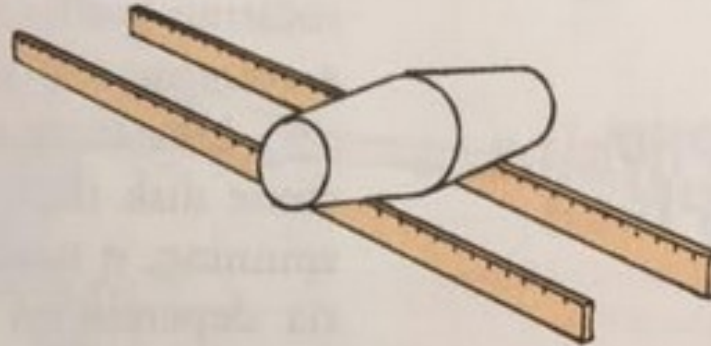


FIGURE 8.5

A pair of fastened cups stay on the tracks as they roll because, when they roll off center, the different tangential speeds due to the taper cause them to self-correct toward the center of the track.

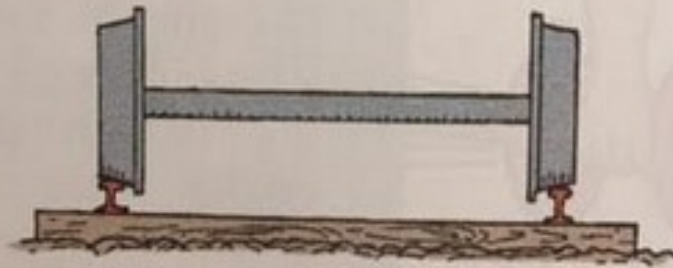


FIGURE 8.6

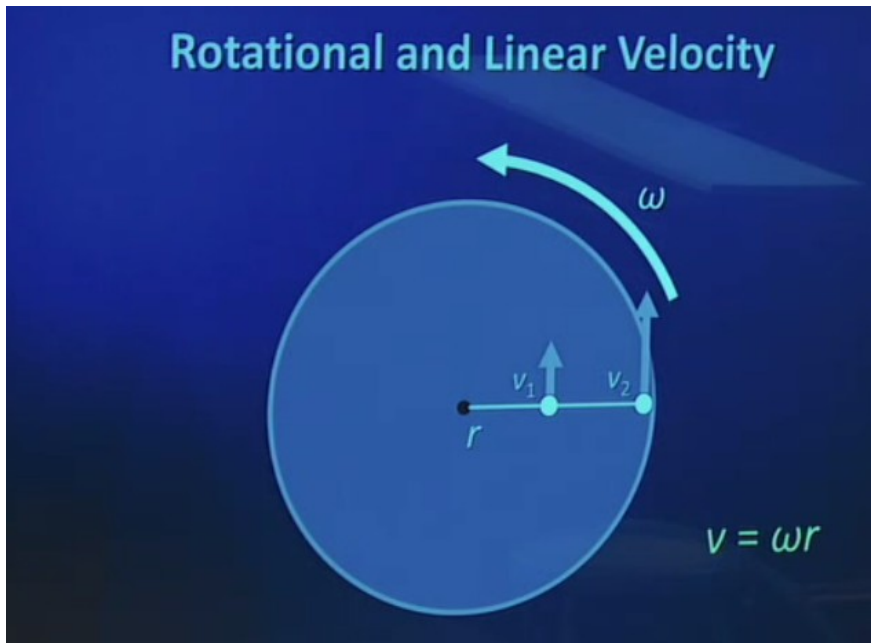
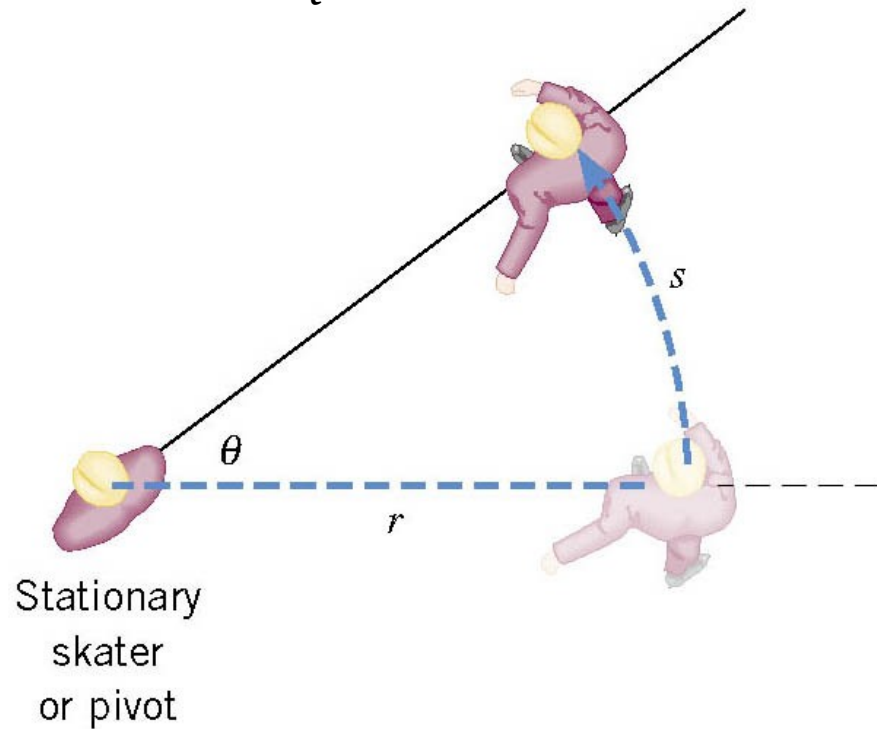
Wheels of a railroad train are slightly tapered (shown exaggerated here).

8.4 Angular Variables and Tangential Variables

$$v_T = \frac{s}{t} = \frac{r\theta}{t} = r \left(\frac{\theta}{t} \right)$$

$$\omega = \frac{\theta}{t}$$

$$v_T = r\omega \quad (\omega \text{ in rad/s})$$



8.4 Angular Variables and Tangential Variables

$$\alpha = \frac{\omega - \omega_o}{t}$$

$$a_T = \frac{v_T - v_{To}}{t} = \frac{(r\omega) - (r\omega_o)}{t} = r \frac{\omega - \omega_o}{t}$$

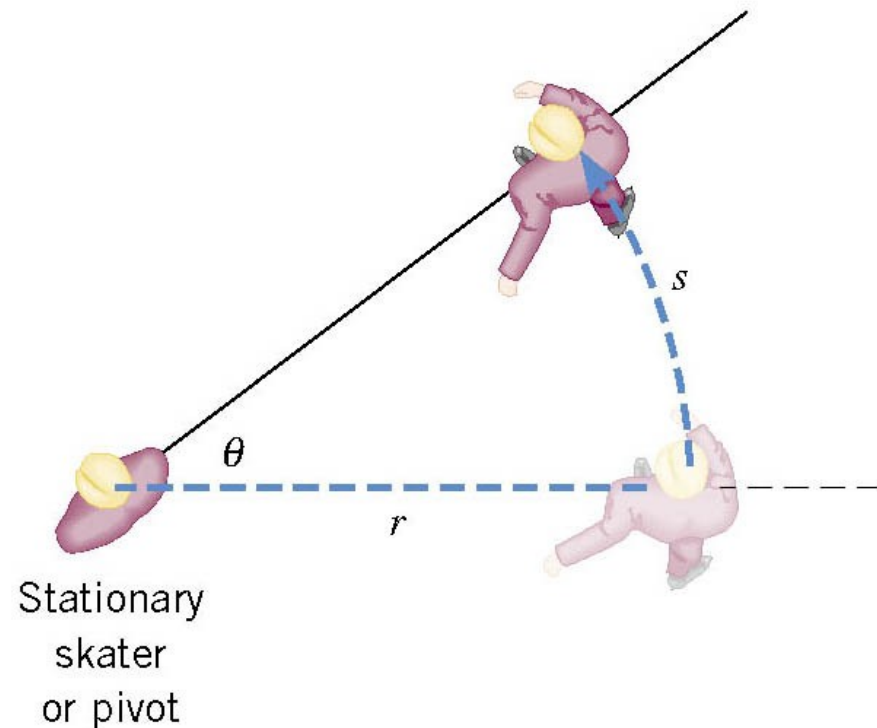
$$a_T = r\alpha \quad (\alpha \text{ in rad/s}^2)$$

So $d = R (\Theta)$ (m $\rightarrow$ rad)

$V = R \omega$ (m/s $\rightarrow$ rad/s)

$a = R \alpha$ (m/s/s $\rightarrow$ rad/s/s)

2pi rad is 1 revolution is 360°

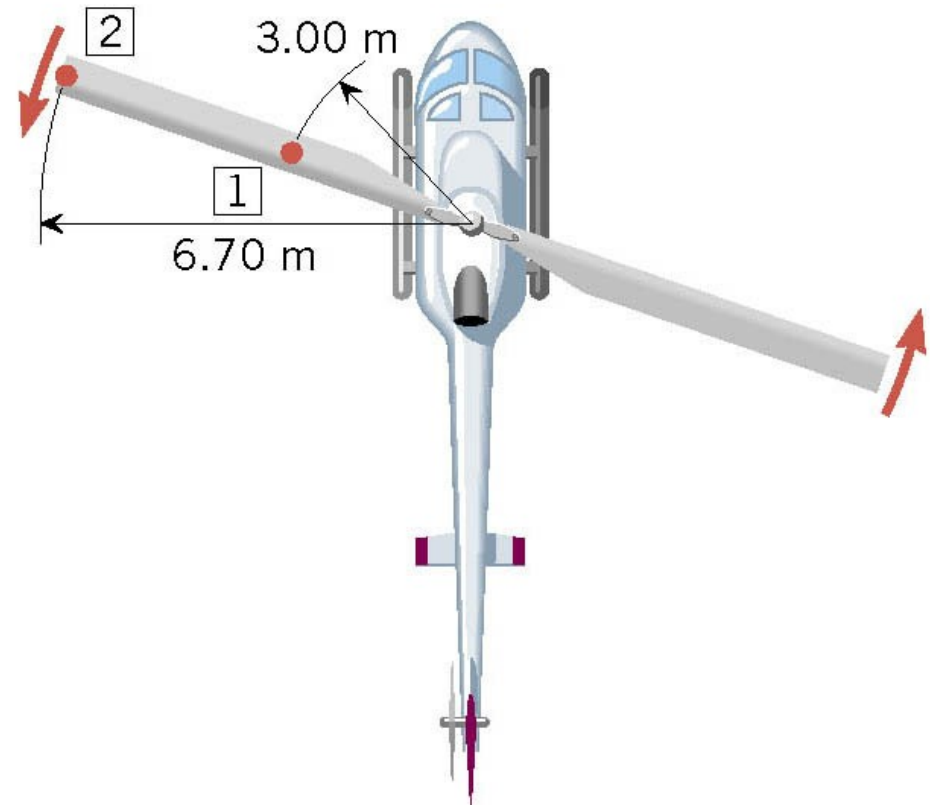


8.4 Angular Variables and Tangential Variables

Example 6 A Helicopter Blade

A helicopter blade has an angular speed of 6.50 rev/s and an angular acceleration of 1.30 rev/s^2 .

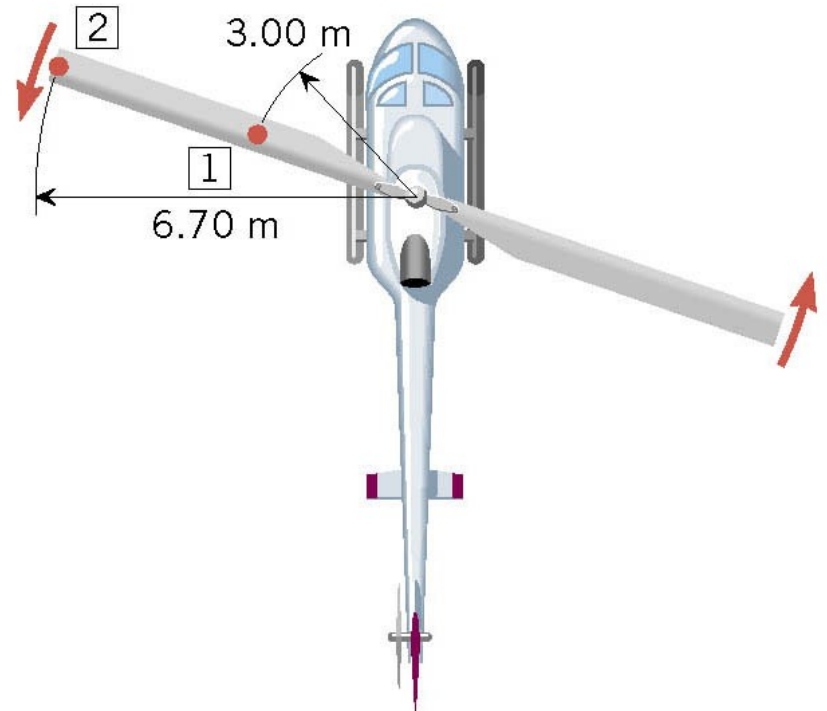
For point 1 on the blade, find the magnitude of (a) the tangential speed and (b) the tangential acceleration.



8.4 Angular Variables and Tangential Variables

$$\omega = \left(6.50 \frac{\text{rev}}{\text{s}} \right) \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) = 40.8 \text{ rad/s}$$

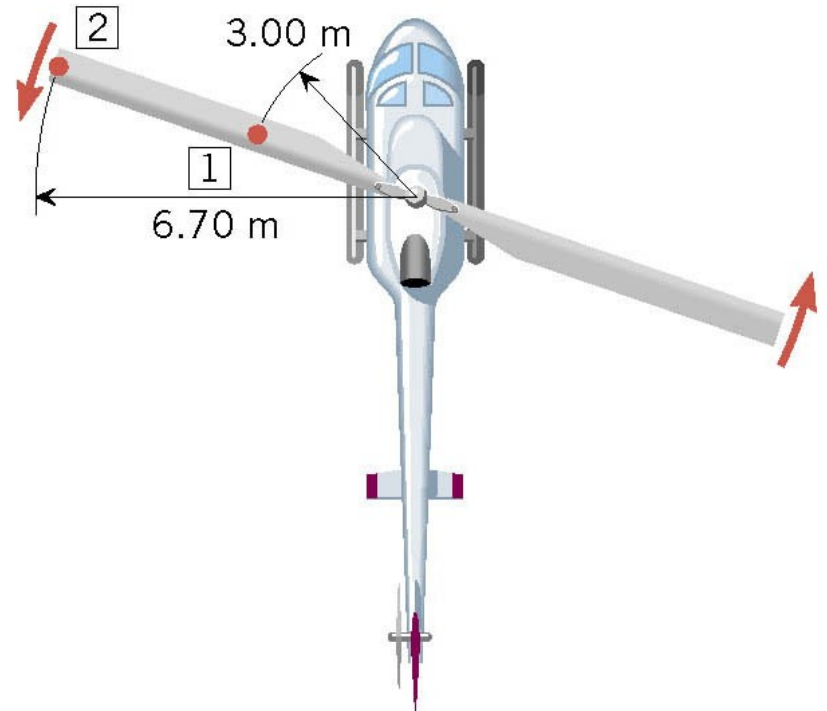
$$v_T = r\omega = (3.00 \text{ m})(40.8 \text{ rad/s}) = 122 \text{ m/s}$$



8.4 Angular Variables and Tangential Variables

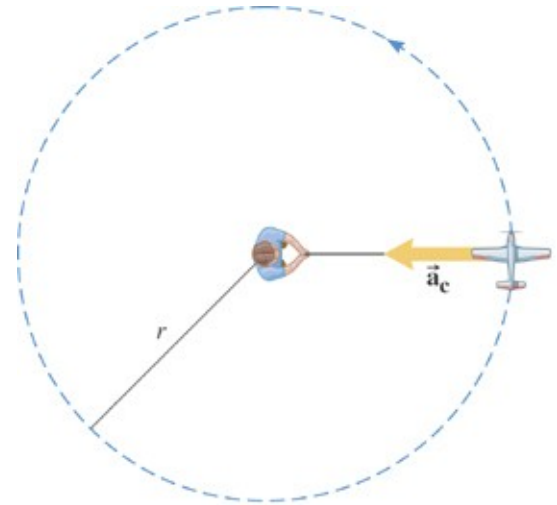
$$\alpha = \left(1.30 \frac{\text{rev}}{\text{s}^2} \right) \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) = 8.17 \text{ rad/s}^2$$

$$a_T = r\alpha = (3.00 \text{ m}) (8.17 \text{ rad/s}^2) = 24.5 \text{ m/s}^2$$

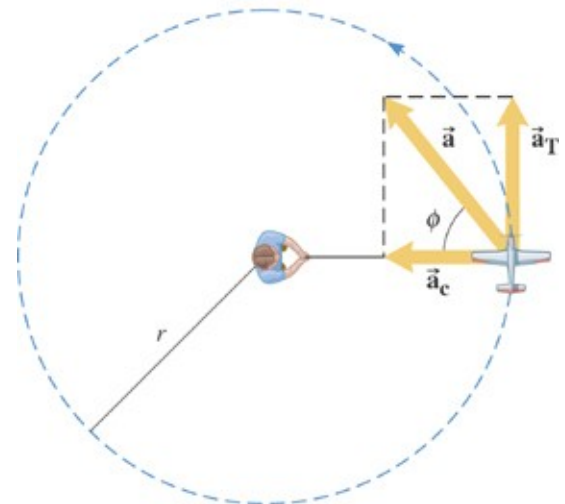


8.5 Centripetal Acceleration and Tangential Acceleration

$$a_c = \frac{v_T^2}{r} = \frac{(r\omega)^2}{r} = r\omega^2 \quad (\omega \text{ in rad/s})$$



(a) Uniform circular motion



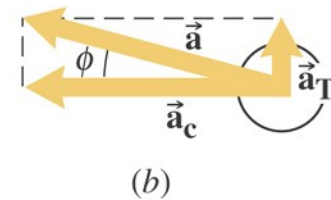
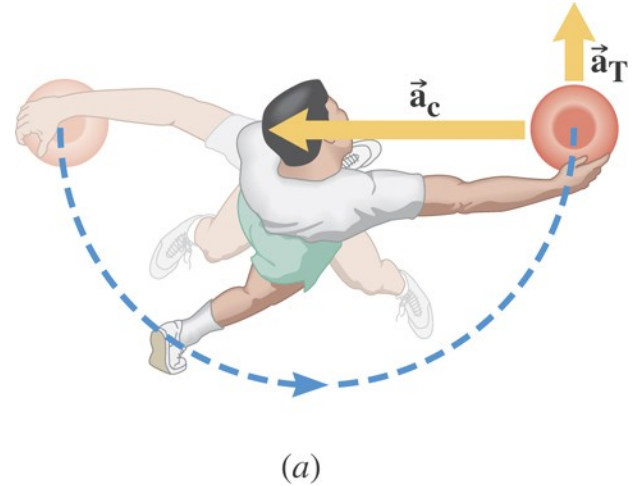
(b) Nonuniform circular motion

8.5 Centripetal Acceleration and Tangential Acceleration

Example 7 A Discus Thrower

Starting from rest, the thrower accelerates the discus to a final angular speed of $+15.0 \text{ rad/s}$ in a time of 0.270 s before releasing it. During the acceleration, the discus moves in a circular arc of radius 0.810 m .

Find the magnitude of the total acceleration.



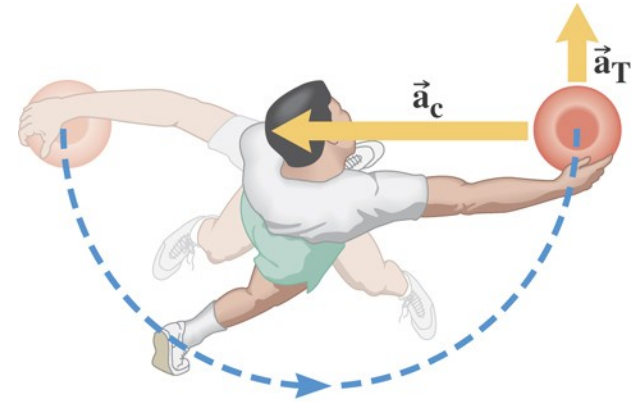
8.5 Centripetal Acceleration and Tangential Acceleration

$$a_c = r\omega^2 = (0.810 \text{ m})(15.0 \text{ rad/s})^2$$

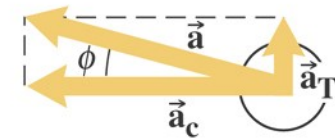
$$= 182 \text{ m/s}^2$$

$$a_T = r\alpha = r \frac{\omega - \omega_0}{t} = (0.810 \text{ m}) \left(\frac{15.0 \text{ rad/s}}{0.270 \text{ s}} \right)$$

$$= 45.0 \text{ m/s}^2$$



(a)



(b)

$$a = \sqrt{a_T^2 + a_c^2} = \sqrt{(182 \text{ m/s}^2)^2 + (45.0 \text{ m/s}^2)^2} = 187 \text{ m/s}^2$$

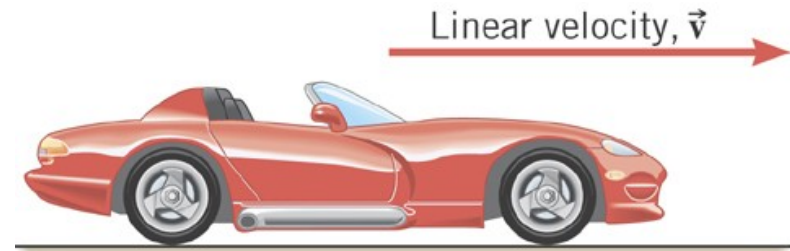
8.6 Rolling Motion

The tangential speed of a point on the outer edge of the tire is equal to the speed of the car over the ground.

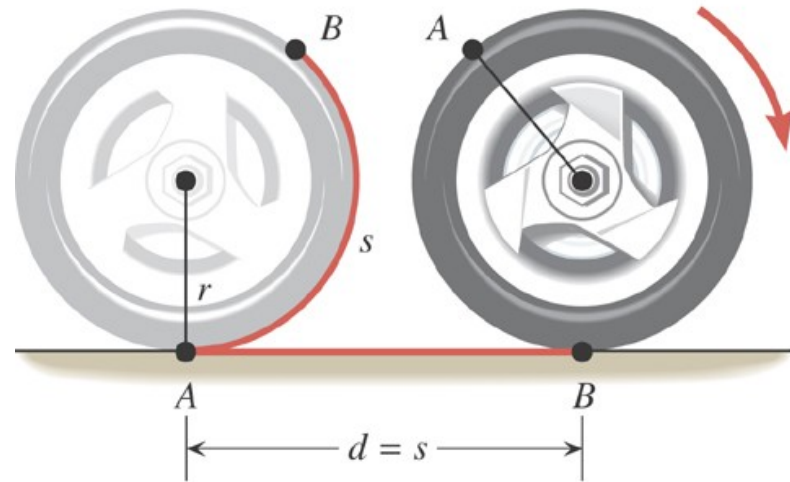
$$v = r\omega$$

$$a = r\alpha$$

Video: rolling without slipping
Tape → watch on your own.



(a)



(b)

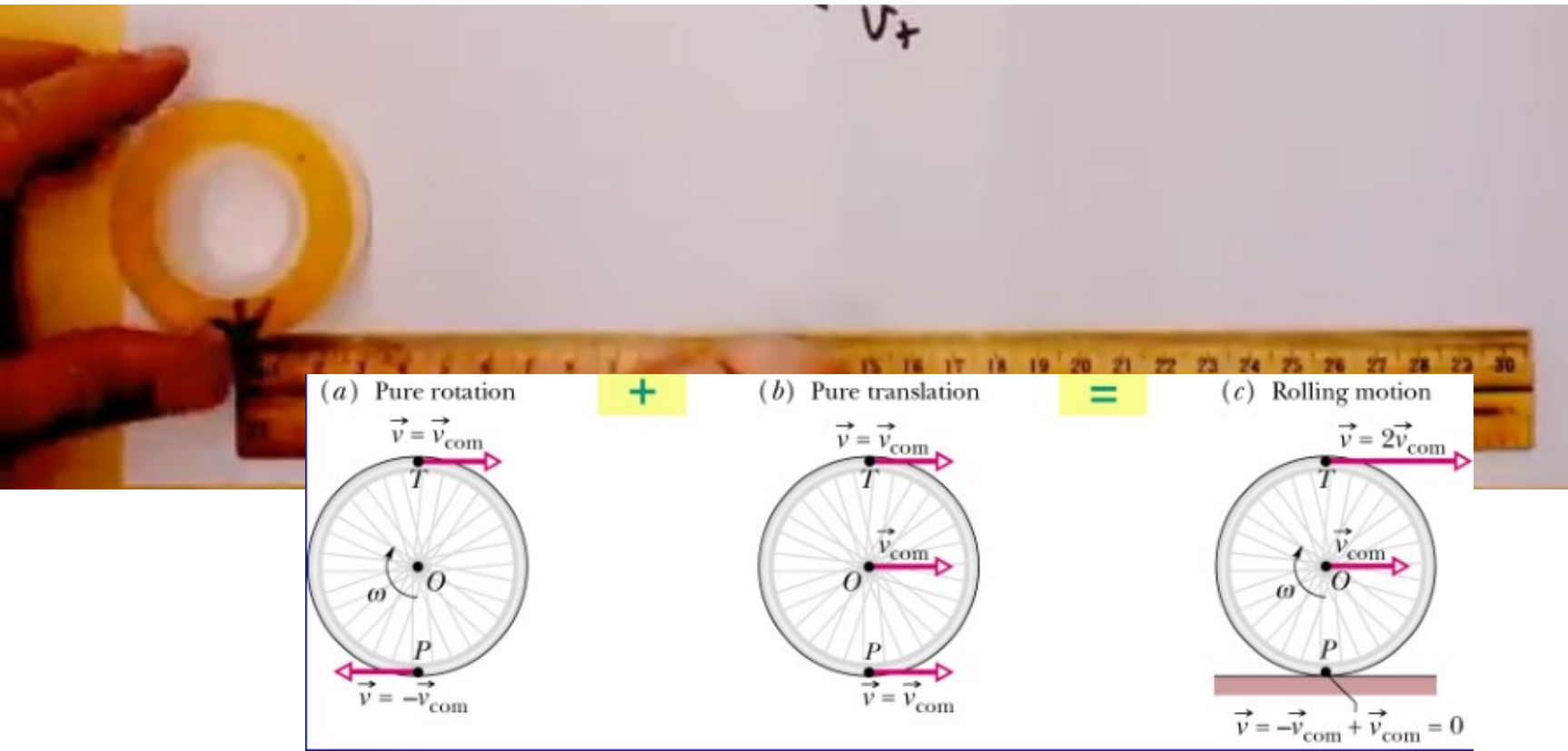
<https://www.youtube.com/watch?v=ZOynwQ75pms>

Watch the video rolling without slipping/

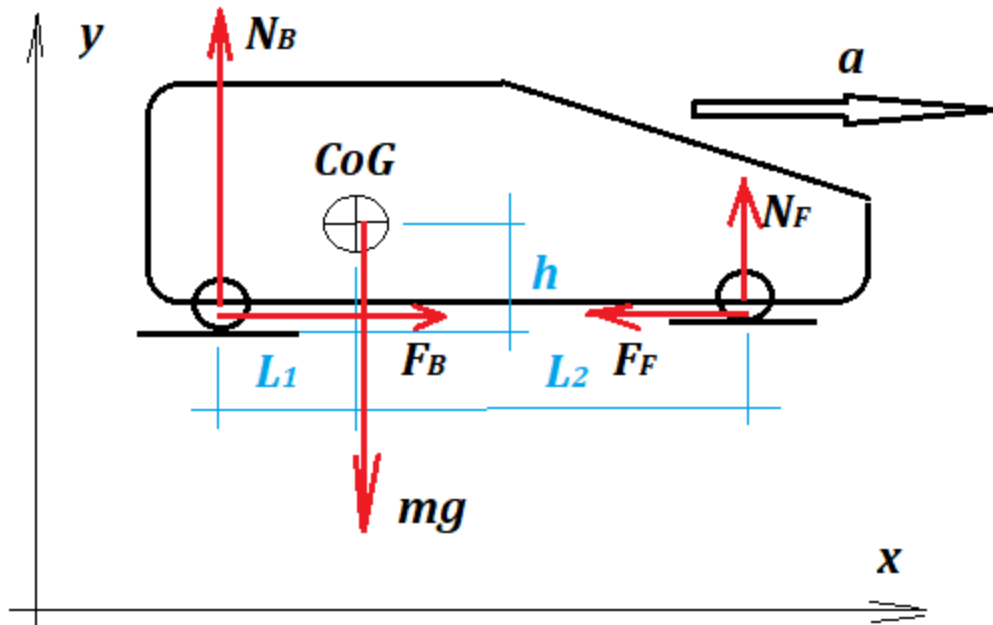
The tape rolls (like the tire of a car) along the ruler for one revolution at a constant rate. The center of mass CM undergoes a translation
And the tape (wheel) a rotation about the center of mass.

The angular displacement of the “wheel” is 2π and the center of mass covers $2\pi R$ So Velocity CM = $V = 2\pi R / T$ and rotational velocity wheel $\omega = 2\pi / T$

So **$V = R \omega \rightarrow$ condition for rolling without slipping**

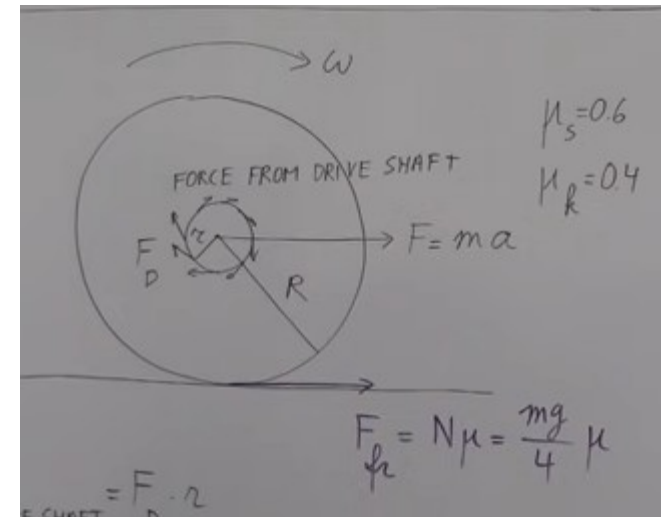


Rear Wheel Drive



<https://www.youtube.com/watch?v=RHYHtfyamOw>

<https://www.youtube.com/watch?v=MYJMGIScxaQ> →

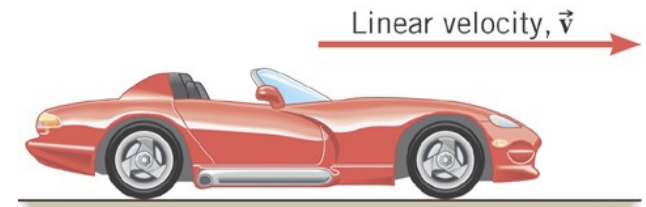


8.6 Rolling Motion

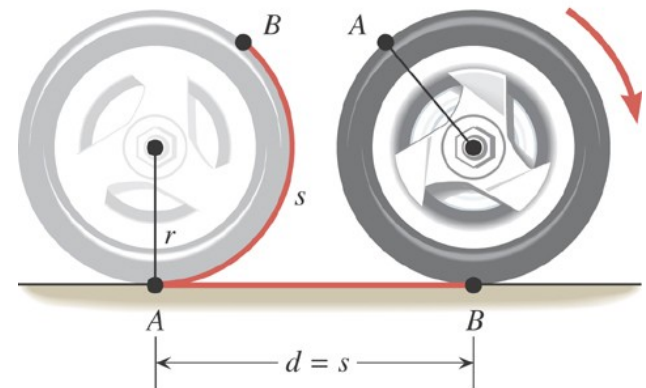
Example 8 An Accelerating Car

Starting from rest, the car accelerates for 20.0 s with a constant linear acceleration of 0.800 m/s^2 . The radius of the tires is 0.330 m .

What is the angle through which each wheel has rotated?



(a)



(b)

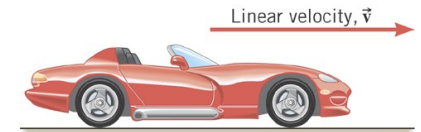
8.6 Rolling Motion

$$\alpha = \frac{a}{r} = \frac{0.800 \text{ m/s}^2}{0.330 \text{ m}} = 2.42 \text{ rad/s}^2$$

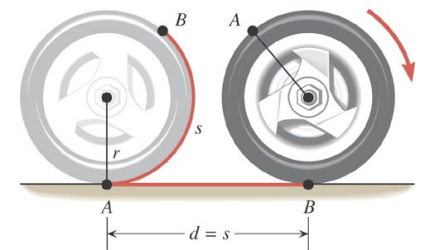
θ	α	ω	ω_o	t
?	-2.42 rad/s^2		0 rad/s	20.0 s

$$\theta = \omega_o t + \frac{1}{2} \alpha t^2$$

$$\theta = \frac{1}{2} (-2.42 \text{ rad/s}^2) (20.0 \text{ s})^2 = -484 \text{ rad}$$



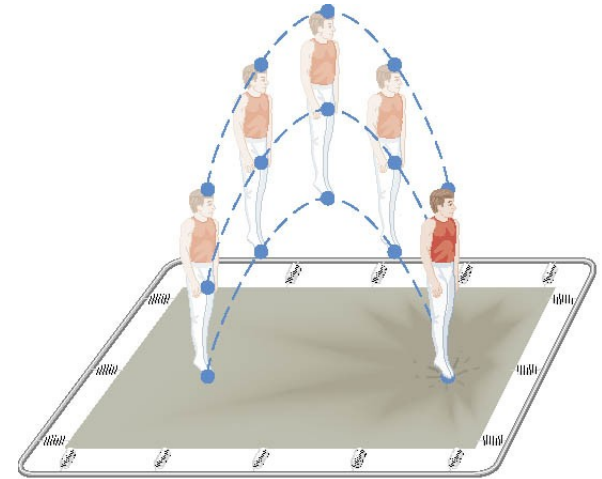
(a)



(b)

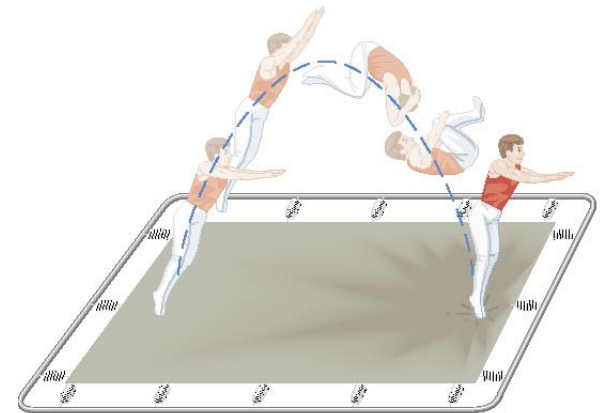
9.1 *The Action of Forces and Torques on Rigid Objects*

In pure translational motion, all points on an object travel on parallel paths.



(a) Translation

The most general motion is a combination of translation and rotation.

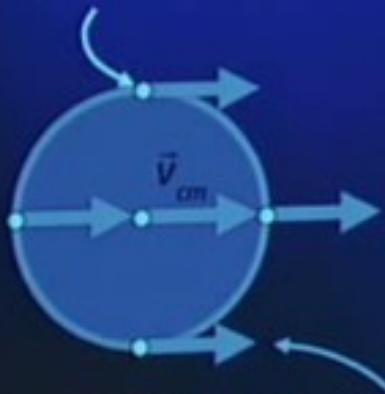


(b) Combined translation and rotation

Rolling:

A Combination of Translational and Rotational Motion

Motion of the CM
plus...



...motion about the CM
equals...

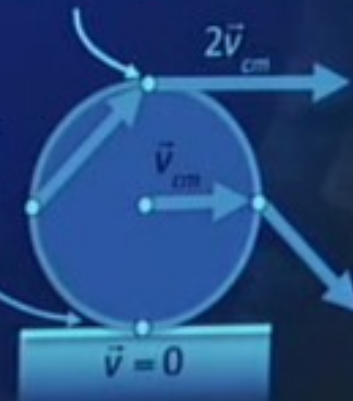


+

=

...motion of individual
points on the wheel.

The
bottom of
the wheel
is at rest!
But only
for an
instant.



These two velocity vectors sum to
give zero velocity at bottom.